ARTICLE IN PRESS

Transportation Research Part D xxx (2017) xxx–xxx

Contents lists available at ScienceDirect

Transportation Research Part D

journal homepage: www.elsevier.com/locate/trd

Editorial

# Land use and transportation in China

In a 2013 paper, Dr. Robert Cervero states that "whatever progress [developed countries] make in reducing greenhouse gas emissions (GHG) and fuel consumption will be quickly eclipsed if rapidly growing countries like India, China, and Brazil continue to mimic American-style patterns of suburbanization, car ownership, and travel" (Cervero, 2013, p. 7).

China, the most populous country, experienced unprecedented urbanization in the past decades and will continue to do so in the near future. The National New-Type Urbanization Plan (2014–2020) aims to move 100 million rural residents to urban areas by 2020.[1] The rural-to urban migration imposes great carbon emissions challenges (Qu et al., 2015; Zhang et al., 2016). How to accommodate the massive number of migrants is an imminent practical problem for Chinese metropolitan areas. Urban sprawl, a hallmark of the US urban development, emerged in urban China (Gao et al., 2016) and to counter it land use policies and plans are being implemented. During the past decade, as well, auto ownership exploded as disposable income increased and national economic policies shifted.[2] Auto-oriented policies gradually dominated transportation decision-making. As congestion deteriorates, many Chinese cities adopt radical policies (such as, license plate auction and road space rationing) to reduce car dependence. Will these land use and transportation policies slow down the enormous impacts of urbanization and motorization and ultimately help China achieve sustainable development?

The literature offers developing countries limited clues, particularly China. Although many scholars investigate the relationships between land use and transportation in developed countries, the lessons are not likely generalizable to China because of its unique characteristics (Feng et al., 2013; Wang and Lin, 2014). For example, Chinese are mainly nurtured under Confucianism; China is a socialist country with strong centralized planning; the stage of land use and transport development in China differs from developed countries. By contrast, international mainstream journals publish limited China-specific studies, particularly those conducted by scholars in mainland China (Cao, 2015; Wang and Zhou, 2017). Rigorous studies published in Chinese journals are also inadequate leaving a large gap for land use and transportation policy-making in China.

The key purpose of this special issue is twofold: (1) to facilitate land use and transportation research in the Chinese context and (2) to generate knowledge to promote evidence-based policy-making in China. Its aim is to attract international scholars to study China and other developing countries, and to bridge Chinese scholars and international scholars for research collaboration, to ultimately enhance policy-making in China.

This special issue attracted almost 50 abstracts. Most were presented at the 9th International Association for China Planning (IACP) Annual Conference at Chongqing University in June 2015, which I was honored to chair. After the standard review process, this issue ends up with 15 papers. Most of them are collaborative studies between Chinese scholars and international scholars (including Chinese abroad). The issue covers a variety of topics, such as land use and travel behavior, residential self-selection, subway investment and travel, license plate restriction, travel impacts of developing new cities, carbon emissions, travel satisfaction, and special populations. It also features two review articles by leading scholars from China. This issue will guide planning research and practice in China for years to come. A brief summary of the studies follows.

License plate restriction (LPR), also called road space rationing, limits vehicle access to a road network based on the numbers of license plates. Several cities in Latin America and South America use LPR to mitigate air pollution and traffic congestion (Han et al., 2010). To reach similar goals, more than 10 Chinese cities adopted this type of policy by 2015. Because it is controversial to prohibit vehicle access, it is important to assess driver acceptance of LPR. Jia et al. (2017) find that 70% of their respondents supported the LPR policy in Tianjin. Model results show that public acceptance is associated with the

[1] http://news.xinhuanet.com/english/china/2014-03/16/c_133190495.htm, accessed on January 23, 2017.
[2] http://news.xinhuanet.com/english/2016-01/25/c_135043964.htm, accessed on January 23, 2017.

http://dx.doi.org/10.1016/j.trd.2017.02.007
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following dimensions: effectiveness of the policy; social norms; awareness of individual responsibility; severity of traffic and related problems; and, advantages and disadvantages of the policy. Further, drivers who accepted LPR are more likely to comply with the LPR policy on restricted days than those with low acceptance; the former were less likely to switch to private vehicles on unrestricted days than the latter. Therefore, public acceptance is critical for the success of LPR.

Although many studies explore the influence of built environment elements on commuting behavior, they often focus on residential neighborhoods elements overlooking those at workplaces (Ding et al., 2014). Using data from Shanghai and a discrete-continuous copula-based model, Sun et al. (2017) investigate the influence of built environment variables at both residential neighborhoods and job locations on commute mode choice and commute distance jointly. They find that the built environment at the residence has a larger impact on commuting behavior that at the workplace. They further conclude that compact and pedestrian-friendly development promotes the use of alternative modes and reduces commute distance.

Developing new cities (or city districts) is the hallmark of urbanization in China. More than 3500 new cities were planned by May 2016 that can accommodate approximately 3.4 billion people.[3] To stimulate development of new cities, many cities relocated city governments to new cities at the urban periphery. Yang et al. (2017) examine how government job relocation (GJR) influences commute mode choice of government employees, using stated preference and revealed preference data in Kunming. They find that cars substitute for alternative means of transport, and that mode choice is dominated by travel time and home location after the GJR is implemented. Based on these results, they question the claimed transportation benefits of job decentralization. They also find that stated preference data and revealed preference data produce different results, which threatens the validity of stated preference analysis.

Although many studies explore the relationship between rail transit and travel behavior using cross-sectional data (Cao and Schoner, 2014; Chatman, 2013), Wu and Hong (2017) employ repeated cross-sectional data to examine the expansion of the subway (metro transit) system on commute mode choice. With four subway lines in operation, four new lines commenced during October 2007 and February 2009 in Beijing. After modeling the data from two individual surveys conducted in 2005 and 2009, they find that subway network expansion increases subway use, but reduces the use of bus and non-motorized transportation and does not affect car use. These findings question environmental benefits of subway expansion. This study also finds that compared to those living in the same neighborhoods, movers tend to use the subway rather than the bus and non-motorized transportation for commuting. This finding lends some support of residential self-selection: those who prefer subway commuting selectively move to areas with subway improvement and use the subway more often.

A heated debate in the field of land use and travel behavior is whether the observed difference in travel behavior is due to residential choice or travel choice (Bohte et al., 2009; Næss, 2009). Many studies attempt to separate the effect of residential self-selection from the built environment effect (Cao, 2014). However, few consider residential self-selection in China (Wang and Lin, 2014). Cao and Yang (2017) examine the influence of built environment elements on carbon emissions (an outcome of travel behavior) in Guangzhou, using structural equations models and statistical controlling for attitudinal factors. They find evidence of self-selection effects. After controlling for residential self-selection, the built environment has separate effects on carbon emissions, which lend support to the implementation of built environment interventions.

The number of vehicles proliferated in China during the past decade. At the same time, Chinese governments made enormous investments on rail transit in the hope of curbing auto ownership and use (Lu et al., 2016). However, few studies examine the impacts of rail transit on auto ownership in China (Huang et al., 2016). Following Cao and Cao (2014) and Huang et al. (2017) employ a case-control design to assess how metro transit affects auto ownership in Xi'an, controlling for built environment elements, attitudes and demographics (the two sources of residential self-selection Mokhtarian and Cao, 2008). Both cross-sectional and quasi-longitudinal analyses show that metro transit development is negatively associated with auto ownership. Since auto ownership is growing in China, it appears that metro transit slows auto ownership growth, instead of reducing auto ownership. This study also assesses their impact on other mobility instruments, including bikes, e-bikes, and motorcycles.

Zhang et al. (2017) also explores the influence of metro transit (subway) on auto ownership, as well as auto use in the form of fuel consumption. This study is unique because its sample includes residents living in resettlement and reformed houses, who have limits in choosing where to live. This subsample of residents becomes a natural experiment to disentangle the effect of subway and residential self-selection than observational studies. They conclude that there is a residential self-selection effect and proximity to the subway reduces car dependency in Beijing. Further, the location of subway stations moderates the effects of subway proximity on auto ownership and fuel consumption.

Jiang et al. (2017a) explore auto ownership and use from the perspective of anti-urban sprawl. In particular, they examine the impacts of residential location, residential neighborhood design, and street network design on the number of cars and car travel distance in Jinan. Following Zegras (2010), they use instrumental variables to predict car ownership, and then incorporate the predicted ownership into the travel distance model. They conclude that limited parking supply, land use mix, pedestrian-oriented street design, and bus rapid transit help reduce resident reliance on personal vehicles. It is worth noting that this study is one of a few that attempt to address the residential self-selection issue by controlling for residential and travel attitudes in China.

User satisfaction is one of the ultimate goals of land use and transportation planning. Few studies explore the relationship between the built environment (an outcome of planning) and travel satisfaction (Cao and Ettema, 2014; De Vos and Witlox,

[3] http://news.xinhuanet.com/politics/2016-07/13/c_1119214482.htm, accessed on January 23, 2017

2016). Ye and Titheridge (2017) investigate the influence of the built environment on commute satisfaction in Xi'an, while controlling for travel attitudes and travel characteristics. Using a structural equations model, they conclude that the built environment affects commute satisfaction through its influence on commute characteristics. They highlight the importance of transit access at residential neighborhood, job-housing balance and green spaces along the commute route. This study also compares and contrasts the results from Xi'an and other cities in developed countries.

Jin et al. (2017) develop integrated land use and transportation models to predict travel demand and associated carbon emissions in the Greater Beijing city region for 2030 and 2050. They examine the following five key land development scenarios: trend, decentralization, densification, spatial mismatch, and targeted major new town. The model results show that the growth in travel demand is mainly attributable to the continuing urbanization, and increasing income and economic development. Among all scenarios, densification seems to have some influence but its effect is very limited. They conclude that a synthesis of sticks and carrots (pricing and land use-transit coordination) could help achieve the goal of sustainable transportation.

TaNa et al. (2017) review the relationships between job-housing balance and commuting in China. They focus on a government intervention unique to China, the Danwei (working units) system, which offers "employees not only with job opportunities but also with a comprehensive package of welfare and services, including housing, medical care, education and pensions." Because of the Danwei reform, job-housing becomes unbalanced and commuting patterns evolve over time: intra-Danwei commuting was dominant before 1978; reverse commuting became prominent in the next twenty years; and long commuting from suburbs to central city has been growing since 1998. They contend that job-housing balance in Danwei compounds reduces commute distance and government intervention could achieve job-housing balance to address the issues related to commuting. They also offer important guidance for future research. This paper is a masterpiece on job-housing balance and commuting in China.

Early interest in the built environment and travel behavior in the US dates back to the late 1970s (Handy et al., 2005). Empirical studies have proliferated since the 1990s. Many studies review the progress in the field (e.g., Cao et al., 2009; Ewing and Cervero, 2010). By contrast, the relationship between land use and travel behavior has not been widely examined in China until a decade ago (Cervero and Day, 2008). With rapid urbanization and automatization, there is a growing interest in using land use and design policies to influence travel behavior in China. Wang and Zhou (2017) review the relevant literature published in both English and Chinese. In particular, they elaborate on the development of the built environment, the evolution of residential environment resulting from changes in housing policy, the unique role of Danwei, the stereotype of neighborhoods, and their effects on and implications for travel behavior. They also offer specific guidance on future research. This paper is a must-read for scholars who are interested in land use and travel behavior research in China.

A growing number of studies address land use and urban transportation in China. However, few pay attention to subgroups of population, such as women, seniors, migrants, and so on (Cao, 2015; Wang and Zhou, 2017). Although there is still a long way to go, a few studies in this special issue help fill the gaps.

Urban villages are an important phenomenon in China's rapid urbanization. They are formed when new development bypasses existing settlements in rural areas to avoid the high costs associated with negotiation with native rural residents. Because of their affordability, urban villages become home to immigrant workers (Hao et al., 2013). On the other hand, due to their underdeveloped infrastructure, they are under enormous demolition pressure (Wu et al., 2013). Using data collected from 12 cities in China, Zhu et al. (2017) investigate whether urban villages offer better job accessibility to the residents than other types of urban settlements. They find that living in urban villages is associated with shorter commute distance and duration. Accordingly, they question the policies of demolishing urban villages. Instead, they propose that governments update the villages and provide affordable housing to accommodate immigrant workers.

As rural workers migrate to urban areas, their expenditures and carbon emissions grow (Zhang et al., 2016). Therefore, residing in cities permanently is likely to increase energy consumption. Using samples from three Chinese cities in 2014 and 2015, Jiang et al. (2017b) explores the factors affecting migrant workers' intent to permanently reside in cities. They find that the intention was influenced by life satisfaction the most, followed by variability in cities and that these two factors are related to the migrants' technical skills, employment status, and income. Further, they conclude that the preference for becoming a permanent resident in cities is not associated with future preference for energy consumption, but barriers and confidence to becoming a permanent resident has effects. That is, stated preference for energy consumption is affected by a migrant's current living and working conditions, especially her social security, medical care and personal health.

While China has undergone significant changes in the built environment, Chinese society is aging. By 2050, 21% of Chinese are projected to be at least 60 years old. How the evolving built environment meets daily needs of the elderly is critical for their quality of life (Curl et al., 2015). Through quantitative analysis and qualitative analysis of the elderly in Nanjing, Feng (2017) finds that seniors in China differ from their counterparts in developed countries in travel behavior and built environment correlates of the behavior, largely because of special social and cultural contexts in China. In particular, transit accessibility, vegetable markets, and open spaces and parks with chess and card rooms affect travel behavior of Chinese seniors whereas the literature suggests auto accessibility, supermarkets and convenience stores, and gyms and sports center are critical for the elderly in developed countries. This study highlights that the land use-travel relationship is context-specific and local knowledge is essential for land use and transportation planning.

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